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CSA5112 - Cryptography and Network Security

CO1 - AT1

Q.1. Invent communication security

- Ceaser cypher

Plaintext : MEET AT NOON

Key = 4

A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13

O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25

Encryption formula : $C = (P+K) \bmod 26$

M	E	E	T	A	T	N	O	O	N
12	4	4	19	0	19	13	14	14	13

Plaintext		Ciphertext	
M	(12+4) mod 26 = 16	16	Q
E	(4+4) mod 26 = 8	8	I
E	(4+4) mod 26 = 8	8	I
T	(19+4) mod 26 = 23	23	X
A	(0+4) mod 26 = 4	04	E
T	(19+4) mod 26 = 23	23	X
N	(13+4) mod 26 = 17	17	R
O	(14+4) mod 26 = 18	18	S
O	(14+4) mod 26 = 18	18	S
N	(13+4) mod 26 = 17	17	R

Encrypted message : QI IX EX RSSR

Decryption :

Decryption formula - P = (C-K) mod 26

Ciphertext		Plaintext	
Q	(16-4) mod 26 = 12	12	M
I	(8-4) mod 26 = 4	4	E
I	(8-4) mod 26 = 4	4	E
X	(23-4) mod 26 = 19	19	T
E	(4-4) mod 26 = 0	0	A
X	(23-4) mod 26 = 19	19	T
R	(17-4) mod 26 = 13	13	N
S	(18-4) mod 26 = 14	14	O
S	(18-4) mod 26 = 14	14	O
R	(17-4) mod 26 = 13	13	N

Decrypted message : MEET AT NOON

Q.2. Intercepted Suspicious Message

Monoalphabetic substitution cypher

Cyphertext : GSRH RH Z HVXIVG

G	S	R	H	R	H	Z	H	V	X	I	V	G
6	18	17	7	17	7	25	7	21	23	8	21	6

Decryption formula : $P = 25 - C$

cipher text		plaintext	
G	25 - 6 = 19	19	T
S	25 - 18 = 7	7	H
R	25 - 17 = 8	8	I
H	25 - 7 = 18	18	S
R	25 - 17 = 8	8	I
H	25 - 7 = 18	18	S
Z	25 - 25 = 0	0	A
H	25 - 7 = 18	18	S
V	25 - 21 = 4	4	E
X	25 - 23 = 2	2	C
I	25 - 8 = 17	17	R
V	25 - 21 = 4	4	E
G	25 - 6 = 19	19	T

GSRH ⟶ THIS

RH ⟶ IS

Z ⟶ A

HVXIVG ⟶ SECRET

∴ Original text = THIS IS A SECRET

b) Justification :

1) Mono alphabetic cypher replaces each letter with another letter using a fixed substitution key.

2) The cyphertext contains words of length 4, 2, 1 and 6.

3) Apply the substitution rule P = 25 - C produces the meaningful sentence "THIS IS A SECRET".

4) The same mapping is used consistently for every occurance of a letter.

Q.3. Secure Messaging System

Vigenere cypher

Plain text : HELLO

keyword : KEY

Encryption formula : C = (P+K) mod 26

7	4	11	11	14
H	E	L	L	O
K	E	Y	K	E
10	4	24	10	4

Position value		Cipher value
H - 7	(7+10) mod 26 = 17	17 (R)
E - 4	(4+4) mod 26 = 8	8 (I)
L - 11	(11+4) mod 26 = 9	9 (J)

Position Value		Cipher Value
L - 11	(11+10) mod 26 = 21	21 (V)
O - 14	(14+4) mod 26 = 18	18 (S)

Encrypted message : RIJVS

b) The influence of repeating keyword is -

1) The keyword is KEY is repeated as KEYKE to match the five plaintext letters.
2) Each keyword letter provides a different numerical shift.
3) The shift is added to the corresponding plaintext value using modulo 26.
4) The repeating keyword determines the sequence of shifts applied to the message.

Therefore, HELLO is encrypted as RIJVS.

Q.4. Suspicious Digital Asset Investigation

Image Steganography

a) Detecting the hidden image data can be done by -

1. File analysis : check the image format, file size, dimensions and metadata for unusual information.
2. File size comparison : compare the suspected image with a genuine copy to identify unexpected size differences
3. Pixel analysis : examine least significant bit (LSB) patterns to identify possible hidden data

4. Metadata inspection : Search for unusual comments, embedded files or extra data appended to the image

5. Steganography tools : Use forensic data tools to detect suspecious patterns and attempt to extract hidden content.

b) Two techniques to strengthen security

1. Encryption before embedding -

encrypt the confidential message using AES before embedding it into the image. If the hidden message is extracted, the attacker still needs the encryption key to read it.

2. Improved LSB embedding

Distribute the hidden data across carefully selected pixels to reduce visible changes and predictable patterns. This can make simple statistical detection more difficult.

Q.5. Legacy encryption in corporate Communication

a) Rail fence transposition cypher :

Ciphertext : WECRLTEERDSOEEFEAOCAIVDEN

Three rail pattern :

Rail 1 : W e c r l t e

Rail 2 : e r d s o e e f e a o c

Rail 3 : a i v d e n

plaintext : we are discovered flee at once

b) This method is vulnerable because

1) Transposition changes the position of letters, does not change the letters themselves.
2) The frequency of each letter remains the same as in the original message.
3) An attacker can test possible arrangements and reconstruct likely plaintexts.
4) It does not provide adequate confidentiality against modern cryptography.

c) The classical transposition method can be replaces with AES-256 in Galois / counter mode (GCM). Use a securely generated encryption key and a unique nonce for each encryption operation.

AES - GCM provides confidentiality and detects unauthorized modification of the encrypted message.

- The receiver can decrypt the ciphertext using the correct key and verify the authentication tag before accepting the message.
- This enhancement provides much stronger protection than a classical transposition cipher against brute force attacks and modern cryptanalysis.
- AES-GCM improves message confidentiality and integrity, making it suitable for secure corporate communication.